

Workforce-constrained maintenance scheduling for military aircraft fleet: a case study

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Abstract The problem is related to a fleet of military aircraft with a certain flying program in which the availability of the aircraft sufficient to meet the flying program is a challenging issue. During the pre- or after-flight inspections, some component failures of the aircraft may be found. In such cases, the aircraft are sent to the repair shop to be scheduled for maintenance jobs, consisting of failure repairs or preventive maintenance tasks. The objective is to schedule the jobs in such a way that sufficient number of aircrafts is available for the next flight programs. The main resource, as well as the main constraint, in the shop is skilled-workforce. The problem is formulated as a mixed-integer mathematical programming model in which the network flow structure is used to simulate the flow of aircraft between missions, hanger and repair shop. The proposed model is solved using the classical Branch-and-Bound method and its performance is verified and analyzed in terms of a number of test problems adopted from the real data. The results empirically supported practical utility of the proposed model.

Keywords Maintenance · Scheduling · Skilled-workforce · Mixed-integer programming · Network flow structure

1 Introduction

The aircraft fleet maintenance plays the most important role to guarantee the safety and reliability of the fleet in commercial airlines and military air forces. In such industries, because of the aircraft complexity and variety, not to mention continuous technological improvements, a broad range of maintenance tasks and high-performance services should be done over the course of a year or even day to keep the fleet availability in a high level. In such situations, we always encounter limited resources such as workforce, facility/equipment capacity, tools, space, and spare parts. Particularly, the workforce

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is considered the highest priority resource, because maintenance tasks are labour intensive, and the workforce performing the tasks is highly-paid and exceptionally skilled in their individual areas. It is estimated that the maintenance function of various airlines accounts for about 10% of their operating costs, about the same as fuel costs (Lam 1995; Cohn and Barnhart 2003). Thus, the available resources should be managed in an efficient manner so that the minimum cost and maximum resource utilization and fleet availability are achieved. As such, the military aircraft MRO (maintenance, repair, and operating equipment) community has recently faced an immediate concern, that is, the reduction in budgets. They found that to achieve reduced total operating costs several changes should take place including a comprehensive revision in maintenance management (Frost & Sullivan Co. at <http://www.frost.com>). An important challenge at the operational level of the maintenance management is the scheduling of maintenance tasks and repair jobs given a short planning horizon. This issue becomes more important in military air forces where we encounter daily missions and high frequency of unexpected faults; whilst, sufficient labour resource should be available simultaneously in both flight line and repair shop; especially, during war time. In this case, Assured Availability and Power-by-the-Hour (Shenneld et al. 2007) are two main elements regularly considered as the aircraft maintenance performance indicators (Beabout 2003).

In this paper, a real maintenance tasks scheduling problem is investigated. The maximization of the availability of an aircraft fleet for a daily pre-scheduled flying program is the ultimate goal. The aircraft are inspected before and after flight and are referred to the repair shop if they have major faults. The maintenance jobs that arrive at the shop should be scheduled in such a way that sufficient aircraft are available for the next planned mission(s). Once the fault has been diagnosed, each job can be completed in a known duration of time and needs a fixed number of skilled technicians to be completed. Thus, the availability of skilled-workforce is the main restriction in the shop. The technicians are assumed to be single-skill or specialized because the internal rules limit them to licence for at most one skill.

A comprehensive review of the literature on maintenance tasks scheduling under the skilled-workforce restriction can be found in Safaei et al. (2011). Although the maintenance tasks scheduling is not a new topic (McCall 1965), just few studies focus on the skilled-workforce availability as main restriction. Wagner et al. (1964) were the first to formulate the PM job-scheduling problem as a mathematical programming model with the aim of minimizing/balancing the fluctuation of workforce requirements during a given horizon. Subsequent studies considered the workforce factor with different assumptions and objectives, including optimal workforce size determination (Vergin 1966), workforce utilization maximization (Roberts and Escudero 1983; Mjema 2002), multi-skilled workforce availability (Gopalakrishnan et al. 1997; Higgins 1998; Ahire et al. 2000; Safaei et al. 2011), subcontracting (Yanga et al. 2003), dynamically assignment of skilled/ranked workforce to machines (Quintana et al. 2009), and workforce cost minimization (Quan et al. 2007; Safaei et al. 2008).

The majority of literature on the maintenance scheduling problem for aircraft fleets addresses the commercial airlines and air cargo fleets (Biró et al. 1992; Clarke et al. 1997; Gopalan and Talluri 1998; El Moudani and Mora-Camino 2000; Sriram and Haghani 2003; Cohn and Barnhart 2003) in which the main restrictions are regularly flight schedule, crew availability, fleet service level as well as maintenance requirements. The commonly-used objectives are operating/system profit maximization (Yan et al. 2006, 2007), turnaround time (Ahire et al. 2000), number of flying hours lost (Boere 1977), or minimization of the workforce size and maximization of the service level (Dijkstra et al. 1994).

To the best of our knowledge there is no prior research work addressing the maintenance tasks scheduling for military aircraft fleet considering the daily missions and fleet availability as a major concerns. In the most relevant studies, Kleeman and Lamont (2005) proposed an evolutionary algorithm to solve the scheduling problem for aircraft engine maintenance in which the goal is to minimize the time needed to return engines to mission capable status and to minimize the associated cost by limiting the number of times an engine has to be taken from active inventory for maintenance. Kozanidis (2009) investigated a bi-objective flight and maintenance planning problem for military air forces with the aim of maximizing two objectives: fleet availability (total number of available aircraft), and total residual flight time (total remaining time that a given aircraft can fly until it has to be grounded for maintenance check). Verma and Ramesh (2007) proposed a multi-objective model to schedule the preventive maintenance tasks in which the fleet reliability, cost and newly introduced criteria, non-concurrence of maintenance periods and maintenance start time factor are simultaneously optimized.

2 Problem definition

Our problem is related to a fleet of military aircraft with a certain flying program to deliver, where ensuring that the availability of sufficient aircraft to meet the flying program is a challenging issue. The Flying Program refers to all the flying activities (namely, waves or sorties) that are planned in a given period (say a day). A flight by any one aircraft is called a *sortie*. More than one aircraft flying together is a *wave*. Throughout this paper, we may use words wave and sortie interchangeably. Each aircraft is inspected before flight (*pre-flight check*) and after landing (*after-flight check*) by technicians. The major failures are referred to the repair shop, and minor faults are fixed whilst the aircraft stays on the flight line. The referred repair jobs are lined up in the shop to await servicing. A number of scheduled preventive maintenance (PM) actions must be accomplished in addition to the unplanned failure repair actions. Generally, whilst flying is underway (and also immediately before and after), the technicians will be divided into three groups for performing the following activities:

- Line Activities (Pre-flight and after-flight checks, including inspection, turn around, re-fuel, top-up fluids, re-role, rearm, etc.)
- Line Rectifications (Minor faults which can be fixed whilst the aircraft stays on the flight line)
- Shed Rectifications (Major faults as well as PM jobs which require work in the repair shop)

When flying is not underway (e.g. during the night shift), the technicians will be in one group repairing the major faults which require work in the shop. As mentioned earlier, serviceable aircraft with minor faults are kept on the flight line. During the day shift when the three groups are in action, technicians are switched between groups as required, with Shed Rectifications taking the lowest priority. The other two groups are equally ranked, with people being placed wherever the chance of getting sufficient aircraft serviceable in time will be maximised. There are usually two, three or four waves during the day of the (say) twelve aircraft on the squadron. Generally, six aircraft are prepared, in the expectation that one or two of these will be found defective during the pre-flight checks and switch-on. The aim is to have maximum required aircraft in each wave.

For illustration, the flight timing for a given individual aircraft is shown in Fig. 1. As shown, the ready aircraft is brought to the line from the hanger, and a pre-flight check is

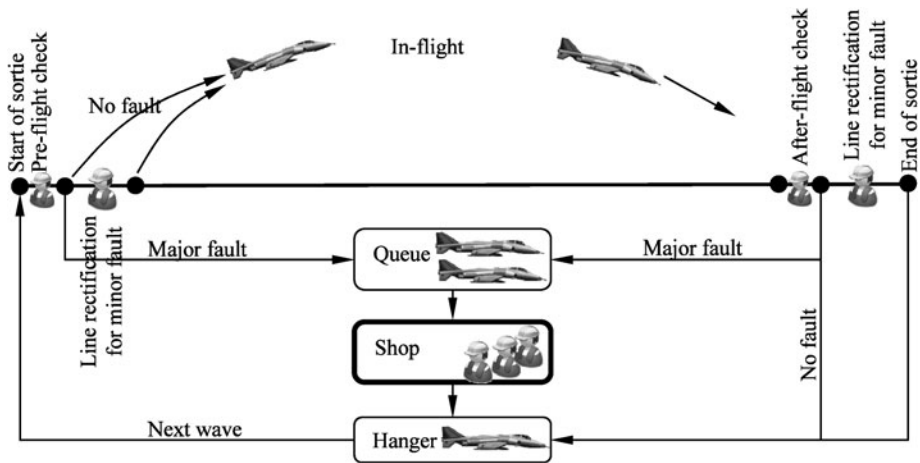


Fig. 1 Flight timing

done, with known duration of time to complete the task. If no fault is detected, the aircraft goes on the mission. If a major fault is detected, it is lined up in the queue to be repaired in the shop. Likewise, once the aircraft finishes the mission, it is moved to the repair shop if a major fault is detected or is repaired on the line if a minor fault is found in the after-flight check. The length of the queue depends on the number of available technicians and the time gap between waves.

Each working day is divided into two shifts, i.e., day shift and night shift. The day shift starts at 8:00 am and ends at 6:00 pm. The night shift starts at 6:00 pm and ends at 8:00 am the next day. The waves are usually scheduled during the day shift. The aircraft with major faults are lined up in the repair shop to be repaired during the day and night shifts. Because each maintenance job means an unserviceable aircraft, the aircraft availability for the incoming flying program directly depends on the throughput and efficiency of the repair shop. Meanwhile, the efficiency of the repair shop depends on the available resources such as workforce, spare parts, tools, space, etc. Since the skilled workforce availability is the most important limitation in the shop, we ignore the availability of other resources to reduce the problem's complexity. Skilled technicians are divided into three trades:

1. Trade 1: Weapons and armament electrical (WP),
2. Trade 2: Airframe mechanical, airframe electrical and propulsion (AF),
3. Trade 3: Avionics/electronics (AV).

The ultimate goal is to schedule the maintenance jobs associated with the major faults in such a way that the fleet availability for the waves is maximised given the skilled workforce limitation. In this study, the *'fleet availability for a given wave'* is defined as the number of fully-mission capable aircraft available to the wave. Hereafter, the term "fleet availability" refers to the mean of fleet availability for all waves as a criterion to measure the fleet availability to the flying program. The detailed discussion will be presented in Sect. 3. It is worth noting that the time to do the line activities and line rectifications (minor faults) is short and those do not affect the performance of the repair shop but rather the efficiency of Shed Rectifications has a direct effect on the resource availability in the flight line. Hence, both line activities and line rectifications are not discussed hereafter and the main focus will be on the Shed Rectifications activity within the repair shop.

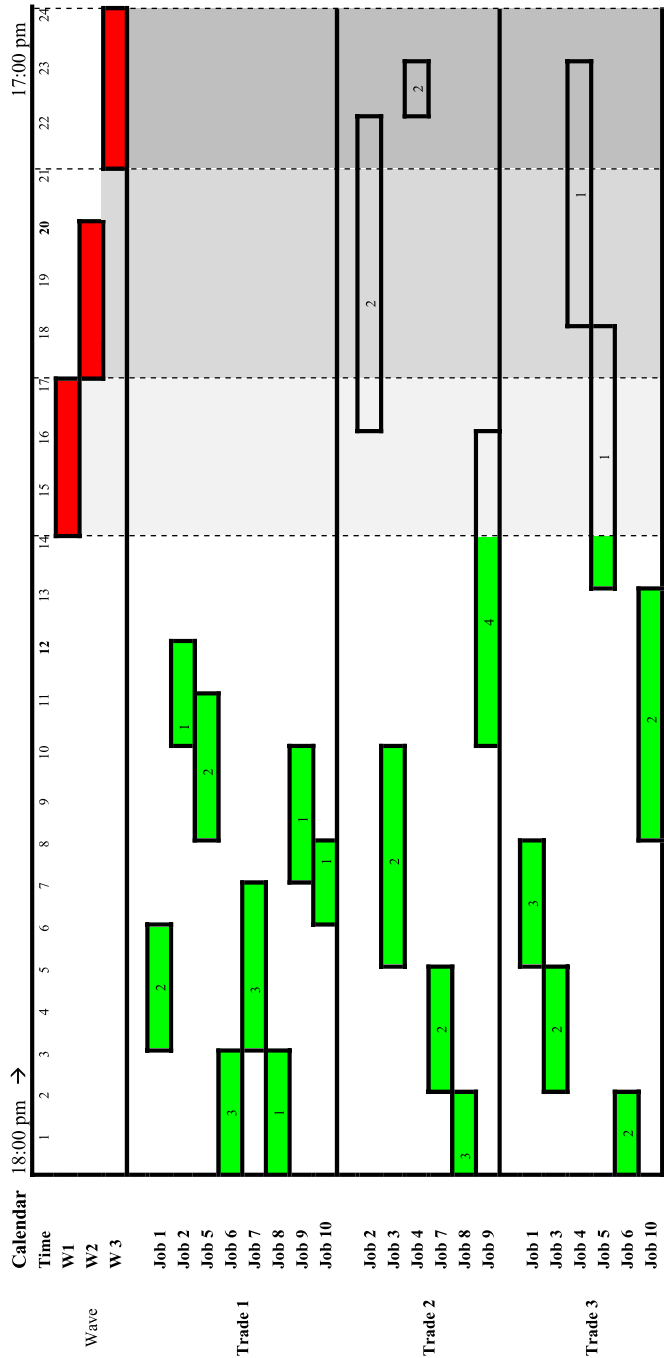


Fig. 2 Typical Scheduling of 10 jobs on 3 trades for a flying program with 3 waves

A perspective of the scheduling of the maintenance jobs during a 24-hour horizon by 3 trades is shown in Fig. 2. There are also three waves scheduled during the day shift and 10 maintenance jobs that must be scheduled during the horizon. Each job requires certain trades so that the duration or processing time of each job by each trade and the required number of technicians of each trade to do the job is known in advance. For example, job 8 needs trades 1 and 2, with durations 3 and 2 hours, and 1 and 3 technicians, respectively. The number within each box indicates the required number of technicians to do the corresponding job. The first three boxes represent the scheduled waves whilst the other boxes represent the maintenance jobs. The dashed lines indicate the start time of the waves as potential deadlines or due dates for the jobs; however, the last one indicates the end of the horizon which can be considered as the start time of a virtual wave.

In the best schedule, we expect that the number of available aircraft immediately before the starting time of the waves will satisfy as much as possible the required number of aircraft. Obviously, to reach a high level of fleet availability, many technicians must work in parallel to complete the jobs in a timely manner, and this means high workforce requirements. Therefore, we have a trade-off between two issues, “*fleet availability*” and “*skilled workforce availability*”.

3 Mathematical formulation

The availability of aircraft for waves can be considered as a network flow structure in which the ready aircraft in hanger or those released from the repair shop are considered as an input flow, and the aircraft failures detected during pre- and after-flight checks are considered as an output flow. In the network structure, waves represent the nodes, with aircraft flowing between them. Without loss of generality, we assume that the waves have been scheduled as a series (i.e., without any time overlap). Hence, the flow network for a given type of aircraft, say k , can be shown by the diagram presented in Fig. 3. E_{wk} represents the expected number of available aircraft type k for wave n and A_k represents the number of ready aircraft type k at the beginning of the horizon.

3.1 Problem assumptions

The assumptions and conditions of the problem are summarised as follows:

1. A 24-hours planning horizon is given beginning of the night shift at 18:00 pm. All maintenance jobs must be accomplished within the upcoming horizon and no job is allowed

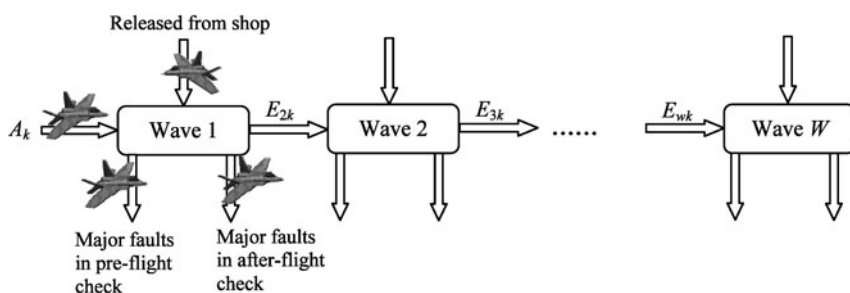


Fig. 3 Flow of aircraft between waves as a network structure

to pull out into the next horizon. The major faults detected during current horizon are lined up in the shop to schedule in succeeding horizon.

2. The starting time and duration of the waves are known in advance, during the incoming planning horizon. Each wave consists of Pre-flight check + in-flight time + after-flight check. All possible minor/major faults are detected during the pre- or after-flight checks. The waves have no any time overlap and those are ordered based on their starting time.
3. A number of maintenance jobs corresponding to the PM tasks, along with the major faults detected during the last horizon, should be scheduled at the beginning of the current horizon. The jobs have the same priority.
4. Because of shortness of the scheduling horizon, rescheduling during the horizon is not possible and the aircraft failed during the current horizon will be repaid in the next horizon.
5. Once an aircraft is released from the repair shop, it is ready for succeeding waves.
6. There is a number of different types of aircraft, each having a number of well-known failure modes extracted from the historical data in the Computerized Maintenance Management System (CMMS) database.
7. The occurrence probability of each failure mode of each aircraft type is known by means of the historical data. Obviously, the probability of major fault detection during after-check is significantly greater than during pre-check because an aircraft in hanger, or released from the shop, or previous wave has been passed successfully a full inspection.
8. The probability of simultaneous occurrence of more than one failure mode for an aircraft is nearly zero.
9. The repair time and workforce requirements for each failure mode are estimated using the CMMS and the expert's knowledge.
10. The time gaps between waves are sufficient to do the inspection, line activities, and minor faults rectification.
11. For the sake of simplicity, the precedence relations between the trades to do the jobs are not considered.

3.2 Notation

Input parameters

T length of the planning horizon (hours)

Aircraft fleet

K number of aircraft types

A_k number of available aircraft type k at the beginning of the horizon (excluding the aircraft in-flight and those in the repair shop before pre-flight checks)

f_k number of possible (or most frequent) failure modes for aircraft type k , where $k = 1, 2, \dots, K$

t_{rik} expected time required to rectify the failure mode i of aircraft type k by trade r , where $i = 1, \dots, f_k$

I_{rik} number of technicians of trade r required to rectify the failure mode i of aircraft type k

Waves

W number of the waves planned/scheduled for the incoming planning horizon

ST_w starting time of the wave w , where $w = 1, \dots, W$

CT_w completion time of wave w

a_{kw} number of aircraft type k required for wave w

Shed rectifications (major Faults)

- M number of jobs lined up in repair shop at the beginning of the planning horizon
- $e_{mik} = 1$ if job m is associated with the failure mode i of aircraft type k ; and equals to 0 otherwise. Note that $\sum_{k=1}^K \sum_{i=1}^{f_k} e_{mik} = 1 \forall m$ means each job is associated with only one failure model of one aircraft
- λ_{rm} number of technicians of trade r required to rectify job m , where

$$\lambda_{mr} = \sum_{k=1}^K \sum_{i=1}^{f_k} I_{rik} e_{mik} \quad \forall m, r$$

- P_{ki}^1 probability that the failure mode i of aircraft type k is detected during the pre-flight check where $\sum_i P_{ki}^1 < 1$. The failure process is the Poisson process with known rate
- P_{ki}^2 probability that the failure mode i of aircraft type k is detected during the after-flight check so that $P_{ki}^2 > P_{ki}^1$
- θ_{ki} probability that the failure mode i of aircraft k is a major fault and has to be repaired in the shop, and $1 - \theta_{ki}$ is the probability that the failure mode i of aircraft k is a minor fault and can be repaired in the line
- ζ_k^1 the overall probability that aircraft type k failed by one of its major faults during the pre-flight check, where $\xi_k^1 = \sum_{i=1}^{f_k} P_{ki}^1 \theta_{ki} \forall k$
- ζ_k^2 the overall probability that aircraft type k failed by one of its major faults during the after-flight check, where $\xi_k^2 = \sum_{i=1}^{f_k} P_{ki}^2 \theta_{ki} \forall k$ and $\zeta_k^2 > \zeta_k^1$

Workforce

- R number of trades (for our case, $R = 3$)
- $\lambda_r^{\max}$ number of technicians of trade r available at the beginning of the horizon. We assume that $\lambda_r^{\max} \geq \max_{1 \leq m \leq M} \{\lambda_{mr}\} \forall r$ which guarantee the existence of the feasible solution.

Decision variables

- Z_{kw} number of aircraft type k assigned to wave w , where $z_{wk} \leq a_{wk}$
- ct_m completion time of job m by trade r
- $C_m^{\max}$ Makespan of job m where $C_m^{\max} = \max_r \{ct_{mr}\}$
- E_{kw} expected number of available aircraft type k for wave w
- λ_r required number of technicians of trade r to do all jobs during the entire horizon.

3.3 Objective function

As noted, the ultimate goal is to maximise the fleet availability for entry waves. The fleet availability for wave w is 100% if the expected number of available aircraft type k at the beginning of wave w , E_{kw} , is greater than or equal to a_{kw} . Hence, considering the demand for different types of aircraft, the mean fleet availability for wave w is defined as $(1/K) \sum_{k=1}^K (z_{kw}/a_{kw})$ where $z_{kw} = \min\{E_{kw}, a_{kw}\}$ represent the number of aircraft type k assigned to wave w . Therefore, the mean fleet availability for all waves as an objective function is expressed as:

$$Z = \frac{1}{W} \sum_{w=1}^W Z_w = \frac{1}{W \times K} \sum_{w=1}^W \sum_{k=1}^K z_{kw}, \quad (1)$$

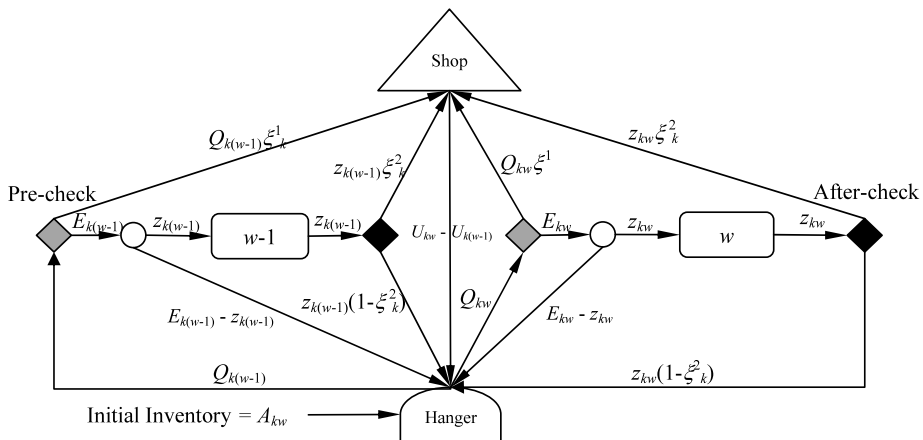


Fig. 4 Simulation of aircraft flow between hanger, shop and two consecutive waves

where $Z_w = (1/K) \sum_{k=1}^K (z_{kw}/a_{kw})$ represents the individual fleet availability for wave w . Whereas the mean fleet availability is our preferable criterion, other similar criteria may be also considered to evaluate the fleet availability to the flying program (please see the [Appendix](#)). The decision variable z_{kw} in (1) directly depends on the expected number of available aircraft type k at the starting time of wave n (ST_n) that is denoted by E_{kn} . Note that the expected number of available aircraft at the beginning of wave is not necessarily the same number of aircraft assigned to the wave. According to the network structure described in Fig. 4, E_{kn} is computed using the following recursive equation:

$$E_{kw} = \{(E_{k(w-1)} - z_{k(w-1)}) + (U_{kw} - U_{k(w-1)}) + z_{k(w-1)}(1 - \xi_k^2)\} \times (1 - \xi_k^1) \quad \forall k, w > 1 \quad (2)$$

where $E_{k1} = (A_k + U_{k1}) \times (1 - \xi_k^1) \forall k$ and $z_{kw} = \min\{E_{kw}, a_{kw}\}$. The key idea behind (2) is that the expected number of aircraft before start time of a given wave directly depends on the number of current available aircraft in hanger, number of aircraft released from the shop so far, and the number of aircraft released from previous wave with no major fault. That is, the first term in (2), i.e., $(E_{k(w-1)} - z_{k(w-1)})$, denotes the expected number of available aircraft within hanger after satisfying the demand of wave $w - 1$. U_{kw} represents the number of aircraft type k released from the repair shop before time ST_w so that,

$$U_{kw} = \sum_{m=1}^M \left(\sum_{i=1}^{f_k} e_{mik} \right) u_{mw}; \quad u_{mw} = \begin{cases} 1 & C_m^{\max} \leq ST_w, \\ 0 & \text{otherwise,} \end{cases} \quad \forall k, w. \quad (3)$$

Thereby, $U_{kw} - U_{k(w-1)}$ is the number of aircraft type k released from the shop between the starting times of waves w and $w - 1$. The term $z_{k(w-1)}(1 - \xi_k^1)$ in (2) indicates the expected number of aircraft type k released from after-flight check of wave $w - 1$ with no major fault. Note that $(1 - \xi_k^1)$ is the probability that aircraft type k takes off without detection of any major fault during the pre-flight check. Finally, in (2), the expected input flow to the wave w , i.e., $Q_{kw} = \{(E_{k(w-1)} - z_{k(w-1)}) + U_{kw} + z_{k(w-1)}(1 - \xi_k^2)\}$ is multiplied by $(1 - \xi_k^1)$ to obtain the expected number of available aircraft type k for wave w , i.e., $E_{kw} = Q_{kw} \times (1 - \xi_k^1)$. The calculation in (2) is schematically shown in Fig. 4 for two consecutive waves $w - 1$ and w , as a detailed extension of Fig. 3.

3.4 Constraints

3.4.1 Regular scheduling constraints

The first series of constraints are classical scheduling constraints to compute the starting and completion times of the maintenance jobs. We use a time-indexed formulation in which the planning horizon is considered as a finite countable set of time slices $t = 1, 2, \dots, T$, where t th time slice $= [t - 1, t)$. The jobs can be started just at the beginning of time slices, i.e., time instances $t = 0, 1, \dots, T - 1$. To implement the formulation, we introduce the decision variable $y_{mkt} = 1$ if the processing of job m is started at time instance t (or equivalently at the beginning of time slice $t - 1$) on trade r ; and equals zero otherwise, where the following constraints must be hold.

$$\sum_{t=0}^{T-p_{mr}} y_{mrt} = 1 \quad \forall m, r; p_{mr} \neq 0, \quad (4)$$

$$ct_{mr} = pt_{mr} + \sum_{t=0}^{T-p_{mr}} t y_{mrt} \quad \forall m, r; p_{mr} \neq 0, \quad (5)$$

$$\begin{cases} C_m^{\max} \geq ct_{mr} & \forall m, r, \\ C_m^{\max} \leq ct_{mr} + M^+(1 - \alpha_{mr}) & \forall m, r, \\ \sum_{m=1}^M \alpha_{mr} = 1 & \forall r, \end{cases} \quad (6)$$

$$C_m^{\max} \leq T \quad \forall m. \quad (7)$$

Equation (4) imposes that the processing of job m on trade r can be started only at one of time-slices over the horizon somewhere within interval $[0, T - p_{mr}]$. Equation (5) determines the completion time of job m by trade r in which $\sum_{t=0}^{T-p_{mr}} t y_{mrt}$ equals the start time of job m on trade r . Notation pt_{mr} in (5) represents the processing time of job m by trade r where $pt_{mr} = \sum_{k=1}^K \sum_{i=1}^{f_k} t_{rik} e_{mik}$. Constraint set (6) computes the Makespan of job m in which the auxiliary variable α_{mr} help us to fully satisfy the equality $C_m^{\max} = \max_r \{ct_{mr}\}$. Inequality (7) ensures that the Makespan of job m cannot exceed the length of the planning horizon based on the first assumption.

3.4.2 Aircraft assignment limitation

As pointed out earlier, the number of aircraft type k assigned to wave n must not exceed the expected available number of aircraft type k at the starting time of wave n (E_{kw}) or the required number of that aircraft type (a_{kw}). In other words, we have

$$z_{kw} = \min \{E_{kw}, a_{kw}\} \quad \forall k, w. \quad (8)$$

3.4.3 Skilled-workforce availability constraint

As pointed out earlier, one major concern in this case study, is the availability of workforce of different skills. That is, the required number of technicians of skill r in each time-slice cannot exceed the available number of technicians of that skill, i.e., $\lambda_r^{\max}$. In other word, assume that $\gamma_{mr}^t = 1$ means job m is being processed by trade r at time-slice t and equals

zero otherwise. Hence, the skilled-workforce availability constraint imposes that

$$\sum_{m=1}^M \gamma_{mr}^t \lambda_{mr} \leq \lambda_r^{\max} \quad \forall r, t.$$

In fact, we encounter a knapsack constraint for each time-slice in which the total required labour of the assigned jobs should not be greater than the available capacity. In accordance with the time-indexed formulation provided in Sect. 3.4.1, the following constraints must be satisfied,

$$\sum_{m=1}^M \left(\sum_{s=\max\{0, t-p_{t_{mr}}\}}^{t-1} y_{mrs} \right) \lambda_{mr} \leq \lambda_r^{\max} \quad \forall r, t, \quad (9)$$

where $\gamma_{mr}^t = (\sum_{s=\max\{0, t-p_{t_{mr}}\}}^{t-1} y_{mrs})$ shows that whether job m is being processed by trade r at time-slice t or not.

3.5 Mathematical model

According to the above discussion, the proposed model is a linear mixed-integer programming written as follows:

$$Z = \frac{1}{W \times K} \sum_{w=1}^W \sum_{k=1}^K \frac{z_{kw}}{a_{kw}} \quad (10)$$

Subject to:

Constraints (4)–(7), (9), and

$$\begin{cases} (C_m^{\max} - ST_w) < (1 - u_{mw})M^+, \\ (C_m^{\max} - ST_w) \geq -u_{mw}M^+, \end{cases} \quad \forall k, w, m, \quad (11)$$

$$\begin{aligned} E_{kw} = & \left\{ (E_{k(w-1)} - z_{k(w-1)}) + \sum_{m=1}^M \left(\sum_{i=1}^{f_k} e_{mik} \right) (u_{kw} - u_{k(w-1)}) \right. \\ & \left. + z_{k(w-1)} (1 - \xi_k^2) \right\} (1 - \xi_k^1) \quad \forall k, w > 1, \end{aligned} \quad (12)$$

$$\begin{cases} z_{kw} \leq E_{kw}, \\ z_{kw} \leq a_{kw}, \end{cases} \quad \forall k, w, \quad (13)$$

$$y_{mrt} \in \{0, 1\}; \quad ct_{mr}, C_m^{\max}, E_{kw} \geq 0.$$

The objective function presented in (10) is to maximise the fleet availability as discussed in Sect. 4.3. Constraint set (11) is to compute u_{km} according to (3). Constraint (12) is the integrated form of (2) based on the relations (2) and (3). Constraint set (13) is used for the linearization of inequality (7).

The proposed model is a time-indexed mixed-integer programming model in which the most of complexity comes from the concave nature of constraints (6) and (11). The unique characteristic of the model is that the objective function does not directly depend on time while the main decision variable is in terms of time. This case is very similar to the scheduling problems with the aim of maximizing the number of early jobs (Hoogeveen et al. 2000)

in which the start time of waves are in fact the potential deadline to the jobs. If a job cannot be completed for a given wave, it might be considered for the succeeding waves. If the maximum fleet availability is achievable with a subset of jobs, the rest will be left out without a certain schedule. Hence, contrary to the classical scheduling problems, the proposed model does not necessarily result in so-called *perfect schedule*. The optimal solution $Y^* = [y_{mrs}^*]$ generated by the proposed model will be *perfect* if:

1. There is no any unnecessary time-gap between the processing of jobs, or equivalently, we have

$$[(st_{mr} = 0) \vee (\exists q; st_{mr} = ct_{qr}) \forall m, r] \quad \text{where } st_{mr} = \sum_{t=0}^T t y_{mrt} \forall m, r$$

is the start time of job m on trade r .

2. The schedule results in the possible best resource utilization by making the resource utilization (i.e., number of used technicians—left side of inequality (9) as close as possible to $\lambda_r^{\max}$. The resource utilization of schedule Y^* still cannot be improved if job m on trade r is found so that its start time can be reduced without violating inequality (9) and any change in Z value.

As pointed out, the model does not take into account the undesirable time gaps and resource underutilization as long as the maximum fleet availability is provided to the waves. The model tries to complete the processing of jobs immediately before waves but no matter how much. We encounter such a situation most likely when nearly 100% fleet availability is simply achievable by current workforce size. That is why we need to employ an auxiliary variable in constraint set (5) to fully satisfy the equality $C_m^{\max} = \max_r \{ct_{mr}\}$ because the considered objective function does not guarantee this equality. In such situations, Y^* should be converted somehow to a perfect schedule. To tackle this issue, Y^* is filtered by a procedure in which the jobs are initially sorted in descending order of start times in Y^* on each trade, i.e., $st_{[1]r} \leq st_{[2]r} \leq \dots \leq st_{[m]r} \forall r$, and then the start time values are adjusted using the dispatching rule $st_{[m]r} = \max_{0 \leq q \leq M} \{ct_{[q]r} | ct_{[q]r} \leq st_{[m]r}; q \neq m\}$; $ct_{[0]r} = 0 \forall m, r$. Note that this filter cannot be always applied; for example, in the case of relaxing Assumption No. 11 we most likely encounter time-gaps in the form of workforce idle-times because of the precedence relations.

4 Experimental results

To verify the performance of the model, some numerical examples adopted from the real data are solved in this section. The model is solved using the classical Branch-and-Bound (B&B) method embedded in LINGO 11.0 software on an x64-based DELL work station with 8 Intel Xeon processor 2.0 GHz and 2 GB memory. Each aircraft type has $f_k = 10$ failure modes divided into three main fault categories WP, AF, and AV similar to the trades. Other detailed fault information is provided in Table 1.

According to the data analysis, the probability of major fault detection during pre-flight check is estimated as $\zeta_k^1 \approx \zeta_k^2/7$. The faults WP, AF and AV arise in each inspection according to the Poisson distribution, with rates 0.42, 2.1, and 2.8, respectively. The values of P_{ki}^2 are obtained in terms of the aforementioned rates. The values of θ_{ki} are obtained based on the discrete scenarios which imply the frequency of referral of a corresponding failure mode to the repair shop as a major fault. The probability of referral of aircraft to the

Table 1 Information related to failure modes of aircraft types

Aircraft	Failure mode		Processing (t_{rik})			Workforce requirements (I_{rik})			Probabilities		
	Category	No. (i)	WP	AF	AV	WP	AF	AV	P_{ki}^2	θ_{ki}^2	ξ_k^2
1	WP	1	3	–	3	2	–	2	0.27	0.76	0.209
	WP	2	3	5	–	2	2	–	0.27	0.45	
	AF	3	2	6	–	2	2	–	0.25	0.75	
	AF	4	3	7	–	2	2	–	0.25	0.57	
	AF	5	–	5	3	–	2	2	0.25	0.03	
	AF	6	–	4	2	–	2	2	0.25	0.98	
	AV	7	–	1	5	–	2	2	0.17	0.25	
	AV	8	–	4	6	–	2	2	0.17	0.93	
	AV	9	3	–	5	2	–	2	0.17	0.6	
	AV	10	–	3	4	–	2	2	0.17	0.12	
2	WP	1	3	–	2	2	–	2	0.27	0.96	0.264
	WP	2	4	3	–	2	2	–	0.27	0.95	
	WP	3	3	2	–	2	2	–	0.27	0.08	
	AF	4	–	6	2	–	2	2	0.25	0.11	
	AF	5	3	6	–	2	2	–	0.25	0.26	
	AF	6	4	7	–	2	2	–	0.25	0.49	
	AV	7	1	–	3	2	–	2	0.17	0.93	
	AV	8	–	4	4	–	2	2	0.17	0.7	
	AV	9	2	–	5	2	–	2	0.17	0.99	
	AV	10	–	1	6	–	2	2	0.17	0.18	

shop during the after-flight check, i.e., ξ_k^2 , is shown in the last column. The time to repair for faults WP, AF and AV follows Log-Normal distribution with parameters (*mean* = 2, *standard deviation* = 3), (4, 10), and (2, 3), respectively. The values of t_{rik} in Table 1 are randomly generated from these distributions. The columns under the “workforce requirements” term show the estimated number of technicians required to rectify the failure modes.

Without loss of generality, it is assumed that all jobs are major faults detected during the flight checks over the last horizon which their information can be extracted from Table 1. The first example consists of $M = 10$ maintenance jobs, $R = 3$ trades including WP, AF, and AV proficiencies, $K = 2$ aircraft types, and a flying program including $W = 3$ waves. The waves have been already scheduled as (8:00 am to 11:00 am), (11:00 am to 14:00 pm), and (15:00 pm to 18:00 pm), respectively. The number of the available technicians in shop at the beginning of the planning horizon is assumed to be four persons for each trade, i.e., $\lambda_r^{\max} = 4 \forall r$. The available number of aircraft types 1 and 2 in hanger at the beginning of horizon are $A_1 = 0, A_2 = 0$. Each wave needs $a_{kn} = 3$ aircraft of each type. Table 2 indicates the job information associated with the first example. As pointed out earlier, each job is associated with a failure mode of an aircraft type. For instance, job 4 is associated with failure mode 7 of aircraft type 1, i.e., $e_{471} = 1$. Thus, the processing time of job 4 by trades AF and AV are computed in constraint (5) as $pt_{42} = \sum_{k=1}^K \sum_{i=1}^{f_k} t_{rik} e_{mik} = t_{271} \times e_{471} = 1$ and $pt_{43} = t_{371} \times e_{471} = 5$ hours respectively.

Table 2 Job information

Job no.	1	2	3	4	5	6	7	8	9	10
Aircraft type (k)	1	1	1	1	1	2	2	2	2	2
Failure mode (i)	1	3	5	7	9	1	2	3	5	9

Table 3 Optimal solution for the first example assuming $\lambda_r^{\max} = 4 \forall r$

Job no.	Start time (st_{mr}) (hrs)			Completion time (ct_{mr}) (hrs)			$C_m^{\max}$
	Trade 1 (WP)	Trade 2 (AF)	Trade 3 (AV)	Trade 1 (WP)	Trade 2 (AF)	Trade 3 (AV)	
1	5	0	0	8	0	3	8
2	3	0	0	5	6	0	6
3	0	7	3	0	12	6	12
4	0	6	6	0	7	11	11
5	0	0	7	3	0	12	12
6	8	0	0	11	0	2	11
7	5	2	0	9	5	0	9
8	2	0	0	5	2	0	5
9	9	5	0	12	11	0	12
10	0	0	2	2	0	7	7
			Turn around $\rightarrow$	12	12	12	12

Table 4 Details of fleet availability

Wave	No. of assigned aircraft types (z_{kw})		Expected No. of aircraft types (E_{kw})		Z_w
	Aircraft type 1	Aircraft type 2	Aircraft type 1	Aircraft type 2	
1	3	3	4.854	4.810	1
2	3	3	4.094	3.864	1
3	3	2	3.363	2.953	0.84

The details of optimal solution are summarized in Tables 3 and 4. The objective function value is $Z = 0.945$, meaning about 94% mean fleet availability for current combination of available skills, i.e., four people in each trade. The individual fleet availability for each wave w , Z_w is shown in the last column of Table 4 where $Z = (1/W) \sum_{k=1}^K Z_w$ according to (1). As is evident in Table 3, all jobs are completed before the starting time of the first wave (i.e., 14 : 00 am) resulting in maximum fleet availability for all waves. In other word, the increase of workforce size of any skill no more improves the objective function value.

To investigate the effect of workforce size on the fleet availability, we perform a sensitivity analysis on $\lambda_r^{\max}$ considering eight different combinations of $\lambda_1^{\max}$, $\lambda_2^{\max}$ and $\lambda_3^{\max}$ where $\lambda_r^{\max} \in \{2, 4\} \forall r$. The obtained optimal solutions are summarised in Tables 5 and 6. As is evident in Table 5, the workforce size significantly affects the fleet availability so that under the worst case scenario (minimum workforce availability, $\lambda_r^{\max} = 2 \forall r$), waves 1, 2 and 3 have a risk of fleet unavailability about 50%, 26% and 26% respectively. Whilst, under the best case scenario ($\lambda_r^{\max} = 4 \forall r$), only the last wave has a risk of fleet unavailability about 26%. However, as an alternative, we can achieve 100% fleet availability for all waves if one

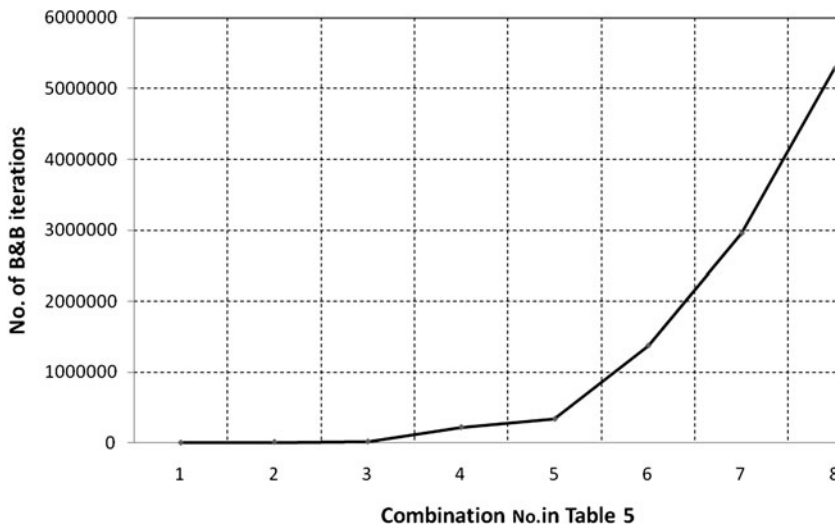
Table 5 Optimal solutions under different combinations of workforce availability

Com. no.	Workforce availability			$C_m^{\max}$ (hrs)										Fleet availability			Objective function	Iter.	CPU time (s)
	$\lambda_1^{\max}$	$\lambda_2^{\max}$	$\lambda_3^{\max}$	1	2	3	4	5	6	7	8	9	10	Z_1	Z_2	Z_3			
8	2	2	2	16	23	13	6	21	23	17	14	12	10	0.5	0.833	0.833	0.722	5284091	1389
7	4	2	2	10	8	23	12	20	8	11	3	18	15	0.666	0.833	0.833	0.777	2955157	577
6	2	4	2	10	11	13	12	23	14	18	11	11	20	0.833	0.833	0.833	0.833	1376457	297
5	2	2	4	6	23	11	12	12	3	21	17	9	14	0.833	1	0.833	0.888	18923	8
4	2	4	4	8	13	13	10	23	5	20	11	16	10	0.833	1	0.833	0.888	11546	7
3	4	2	4	12	9	17	11	12	6	12	3	23	8	1	0.833	0.833	0.888	337895	48
2	4	4	2	13	8	13	10	23	3	4	9	9	18	1	0.833	0.833	0.888	220862	31
1	4	4	4	12	12	11	8	5	12	6	6	11	11	1	1	0.833	0.944	1401	1
	*4	4	4	8	6	12	11	12	11	9	5	12	7	1	1	1	1	1595	1

* 100% fleet availability when $A_1 = 0$ and $A_2 = 1$

Table 6 Details of fleet availability

$\lambda_1^{\max}$	$\lambda_2^{\max}$	$\lambda_3^{\max}$	E_{11}	E_{12}	E_{13}	E_{21}	E_{22}	E_{23}	z_{11}	z_{12}	z_{13}	z_{21}	z_{22}	z_{23}
2	2	2	1.940	2.65	3.13	2.886	3.23	2.34	1	2	3	2	3	2
4	2	2	2.910	2.42	2.91	2.886	3.23	3.30	2	2	2	2	3	3
2	4	2	3.880	3.15	2.45	2.886	2.27	3.60	3	3	2	2	2	3
2	2	4	3.880	3.15	2.45	2.886	3.23	3.30	3	3	2	2	3	3
2	4	4	3.880	3.15	2.45	2.886	3.23	3.30	3	3	2	2	3	3
4	2	4	3.880	4.12	3.39	3.848	2.94	2.32	3	3	3	3	2	2
4	4	2	3.880	3.15	2.45	3.848	2.94	3.28	3	3	2	3	2	3
4	4	4	4.850	4.09	3.36	4.810	3.86	2.95	3	3	3	3	3	2

**Fig. 5** Effect of labour resource availability on B&B's computational effort

aircraft of type 2 is available in hanger at the beginning of horizon, $A_2 = 1$. The findings show that a small change in workforce size results in completely different schedules as can be seen from the values under $C_m^{\max}$ column in Table 5. For example, when $\lambda_1^{\max}$ is increased by 2 at the second row, the schedule obtained for combination (4, 2, 2) significantly differs from the schedule for combination (2, 2, 2) with an average of 8.65 hrs and standard deviation of 5.6 hrs in terms of start time of the jobs. Another interesting finding is high sensitivity of B&B's computational effort to the labour resource availability as shown in couple of last columns of Table 5. That is, the decrease of the workforce size most likely increases progressively the number of iterations of B&B, as depicted in Fig. 5.

The main reason for this behaviour is increasing of the computational effort for the upper bound computation in B&B method. In fact, the smaller the workforce size, the lesser the nodes are fathomed because of existence of a weak upper bound. Important point is that by decreasing the workforce size, the size of feasible space decreases or equivalently the infeasible region grows intensively. In such a case, the finding of a strong upper bound becomes more difficult especially when the penalization strategy is used to generate the

Table 7 Job information—second example

Job no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Aircraft type (k)	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
Failure mode (i)	2	4	6	6	7	9	10	1	2	2	5	5	6	6	9

Table 8 Flying program information—second example

Wave no.	Stating time	Ending time	a_{1w}	a_{2w}
1	6:00 am	9:00 am	4	4
2	9:00 am	12:00 pm	4	4
3	12:00 pm	15:00 pm	4	4
4	15:00 pm	18:00 pm	4	4

upper bound. However, this behaviour is not always true for all instances such as in the second example.

The data associated with the second example consisting of $M = 15$ jobs and a flying program including $W = 4$ waves are provided in Tables 7 and 8. Other parameters are set same as in the first example. The optimal solutions corresponding to the different workforce availability combinations with $\lambda_r^{\max} \in \{4, 6\}$ are summarized in Tables 9 and 10. As depicted in Table 9, while there are no feasible schedules for some combinations, the rests result in nearly same fleet availability but different schedules. If combination (4, 4, 4) is considered as a basis, the increase of workforce size by 2 or 4 persons does not surprisingly improve the mean of fleet availability and therefore the combination (4, 6, 4) is the best choice with lowest workforce size, unless other criteria are considered.

The third example consists of $M = 20$ jobs and a flying program including $W = 4$ waves scheduled based on Table 8, where $a_{kw} = 5 \forall k, w$. The job information is provided in Table 11. The current workforce size is assumed to be (4, 8, 4) that is the worst case scenario for this example. Other parameters are set same as the first example. We could not achieve the optimal solution even after 7 hours runtime; however, the best obtained objective function value is $Z = 0.975$. The individual fleet availability for the waves are $Z_1 = 0.89$, $Z_2 = 1$, and $Z_3 = 1$. The computational effort for this case is 127,177,980 iterations. It is interesting to say that when $\lambda_2^{\max}$ is increased by 2, i.e., combination (4, 10, 4), we reach the 100% fleet availability for all waves while the spent computational effort is reduced to 200490 iterations or about 40 seconds. As a general result, the larger the problem dimension, the more it affects the sensitivity of computational effort to the resource availability. The significant effect of different combinations of workforce availability on the optimal schedule is shown very well in Fig. 6.

The findings reveal that using the classical B&B method, the computational effort does not necessarily depend on the problem dimension but highly depends on the workforce size and combination. Moreover, the runtime is more affected by the number of waves or flying program's demand rather than the number of jobs. Likewise, the generated schedule highly depends on the workforce size and combination. That is, a small change in the workforce size or combination most likely results in a significant change in the schedule.

As a general remark, the proposed model is a proper tool to solve the most of real size instances except very special cases in which the minimum size of workforce is available. Whereas the number of waves is fixed, the proposed model can be also used for these special

Table 9 Optimal solutions under different combinations of workforce availability

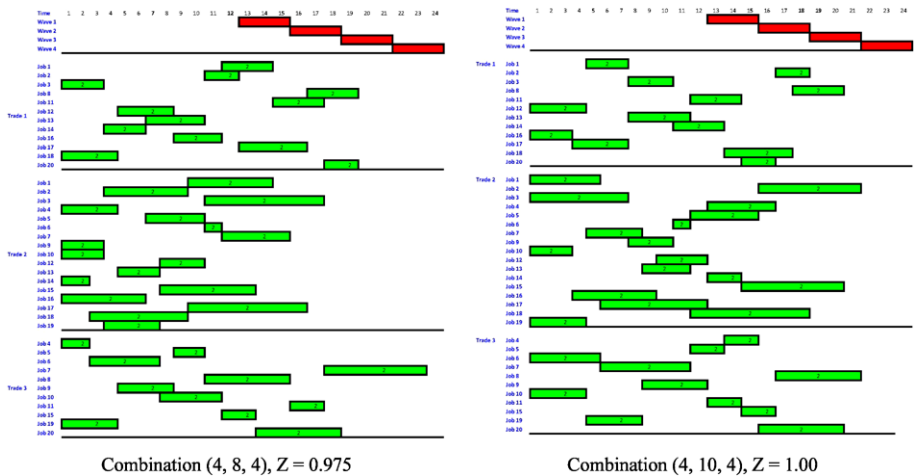
Workforce availability				$C_m^{\max}$ (hrs)																Fleet availability				Objective function	Iter.			
$\lambda_1^{\max}$	$\lambda_2^{\max}$	$\lambda_3^{\max}$		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Z_1	Z_2	Z_3	Z_4						
4	4	4		No feasible solution found																-	-	-	-	-	-	-	-	-
6	4	4		No feasible solution found																-	-	-	-	-	-	-	-	-
4	6	4		9	20	13	4	11	15	6	8	14	11	20	9	7	16	11	0.875	1	1	1	0.96875	17140				
4	4	6		No feasible solution found																-	-	-	-	-	-	-	-	-
6	6	4		20	14	8	4	7	11	17	3	12	9	7	9	19	15	10	0.875	1	1	1	0.96875	5282				
6	4	6		No feasible solution found																-	-	-	-	-	-	-	-	-
4	6	6		12	17	6	20	7	10	10	3	16	7	13	12	9	20	5	1	0.875	1	1	0.96875	25925				
6	6	6		13	14	4	19	8	7	7	6	16	15	12	6	9	21	11	0.875	1	1	1	0.96875	5376				

Table 10 Details of fleet availability

$\lambda_1^{\max}$	$\lambda_2^{\max}$	$\lambda_3^{\max}$	E_{11}	E_{12}	E_{13}	E_{14}	E_{21}	E_{22}	E_{23}	E_{24}	z_{11}	z_{12}	z_{13}	z_{14}	z_{13}	z_{13}	z_{13}
4	4	4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
6	4	4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
4	6	4	3.880	5.093	4.127	4.159	4.810	4.571	4.340	4.119	3	4	4	4	4	4	4
4	4	6	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
6	6	4	3.880	4.123	4.156	4.188	5.772	5.497	4.269	4.050	3	4	4	4	4	4	4
6	4	6	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
4	6	6	4.850	3.891	4.132	4.166	4.810	4.571	4.340	4.119	4	3	4	4	4	4	4
6	6	6	3.880	5.093	4.127	4.159	5.772	5.497	5.231	4.976	3	4	4	4	4	4	4

Table 11 Job information—third example

Job no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aircraft type (k)	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2
Failure mode (i)	2	3	4	6	6	7	8	9	10	10	1	2	2	3	4	5	6	6	8	9

**Fig. 6** Typical Gantt charts associated with third example

cases if we can somehow speed up the solving method. To this end, an ad-hoc heuristic method or a proper Lagrangian relaxation may help to compute the stronger upper bounds and consequently to reduce the computational effort. As an idea, decomposition of the model in terms of skill/trade may result in high-quality upper bounds. Moreover, a homogenous time-dependent objective can be also combined with (10) to obtain the perfect schedules and to reduce the effect of possible degenerate solutions.

5 Conclusion

A real maintenance scheduling problem associated with the operation of a fleet of military aircraft has been investigated in which the availability of the aircraft sufficient to meet the requirements of daily flying program is a challenging issue. The flying program consists of a number of independent missions, each requiring a fixed number of aircraft. The aircraft are inspected before and after flight, and those with major faults are lined up in the shop for repair. The repair shop is responsible to rectify the repair jobs and possible PM tasks in a timely manner to provide the maximum fleet availability for the next flying program. The availability of the skilled-workforce is the major concern in the shop.

The problem is formulated as a time-indexed mixed-integer programming model to maximize the fleet availability whilst considering the skilled-workforce restriction. We use the network structure to simulate the flow of aircraft between the shop, hanger and missions. That is, the missions, hanger and shop are considered as the nodes, with the aircraft flowing between them. The flow equilibrium equations are added to the mathematical model as

constraint. The applicability and performance of the proposed model is verified by a number of real instances under different combinations of workforce sizes. The results indicate that the proposed model can correctly describe the problem and satisfies our expectations in the sense of sensitivity analysis on the workforce availability.

The obtained results show that the computational effort of the solving method and generated schedule highly depends on the workforce size and combination. Moreover, the number of jobs doesn't affect the runtime as much as the number of missions and their demand volume does. These issues may motivate one to develop the improved and enhanced solving methods to do rapid sensitivity analysis as an important necessity.

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Appendix: Alternative Objective Functions

The reason to select the *mean fleet availability* (1) as a criterion to measure the availability of the fleet to the flying program is that each wave, as an individual mission, must be performed even when a small fleet size is available. In other words, we prefer that $z_{kw} > 0 \forall k, w$. Otherwise, other criteria such as *overall fleet availability*

$$Z = \frac{1}{W \times K} \frac{\sum_{w=1}^W \sum_{k=1}^K z_{kw}}{\sum_{w=1}^W \sum_{k=1}^K a_{kw}}$$

may be considered. In that case, it may not be possible to perform some waves due to lack of aircraft, even though a higher level of overall availability may be achieved with respect to mean fleet availability.

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